



Electrolyte System for High Voltage Li-Ion Cells

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Ethylene carbonate is a co-solvent used in virtually every lithium ion cell produced today because it enables operation of both the positive and negative electrodes. Most battery scientists believe ethylene carbonate is essential. Surprisingly, totally removing all ethylene carbonate from typical organic carbonate-based electrolytes and adding small amounts of electrolyte additives creates cells that are better than those containing ethylene carbonate. For example an electrolyte of only 2% vinylene carbonate and 98% ethyl methyl carbonate, with selected additives, provides excellent performance to Li[Ni_{0.4}Mn_{0.4}Co_{0.2}]O₂/graphite cells cycled up to 4.4 V which increases their energy density by at least 10%. The cells have low impedance, low rates of electrolyte oxidation, good graphite passivation, low gas generation, acceptable conductivity and low cost. This discovery opens an entirely new space for electrolyte development.

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Li-ion cells are now used in electric vehicles (EVs) and grid energy storage systems. However their energy density should be increased and their cost should be decreased. Typical layered materials used in commercial cells are not used through their entire potential range of structural stability. For instance, Li[Ni_{0.4}Mn_{0.4}Co_{0.2}]O₂ (NMC(442)) material is used with a potential cutoff of 4.28 V vs. Li/Li⁺ in Li-ion cells despite being structurally stable up to 4.78 V vs. Li/Li⁺.^{1,2} This limitation of the upper cutoff is used in order to limit electrolyte oxidation and polarization growth²⁻⁷ in order to extend cell lifetime.⁸ Figure 1a shows that increasing the potential cutoff of NMC(442) material from the typical 4.28 V to 4.48 V and to 4.78 V vs. Li/Li⁺ leads to an energy density gain (of the positive electrode material) of 18% and 36%, respectively.

Surface coatings,^{6,9-12} electrolyte additives,^{5,7,13-20} or new solvent systems²¹⁻³⁰ have been used with layered materials to be charged and discharged to high potentials. For instance, Figure 1b shows the impact of electrolyte additives in NMC(442)/graphite cells with a typical electrolyte of LiPF₆ dissolved in ethylene carbonate (EC, Figure 1c) and ethyl methyl carbonate (EMC, Figure 1c), cycled to 4.4 V and 4.5 V. Figure 1b clearly shows the performance improvement when vinylene carbonate (VC, Figure 1c) is substituted by propene-1,3-sultone (PES, Figure 1c) in a ternary additive blend. While the improvement in Figure 1b is substantial, it is still not enough for applications such as EVs or grid energy storage.

Successfully operating Li-ion cells to high voltage may require the use of a combination of new solvent blends, additives and coatings. It is generally believed that new solvent blends should have oxidation potentials higher than that of alkyl carbonates (i.e. >5 V vs. Li/Li⁺).³¹ Most proposed solvent blends consist of fluorinated compounds,^{21,22,24,28,32} nitriles or dinitriles,^{33,34} or sulfones.^{25,26} However these new systems can suffer from inadequate wettability of typical olefin separators,²⁶ high viscosity,^{23,26} high gas production and high impedance²¹ or potential cost and safety concerns (e.g. fluorinated solvents).

It is shown here that, surprisingly, EC-free-linear alkyl carbonate-based electrolytes containing small amounts of passivating agents can lead to cells cycled up to 4.4 V with long cycle and calendar life. This electrolyte system is shown to have good conductivity, provide low impedance growth and low levels of electrolyte oxidation during high voltage cycling of NMC(442)/graphite cells.

Experimental

Chemicals.—All chemicals used were obtained from BASF (LiPF₆, purity 99.94%, water content 14 ppm; ethylene carbonate (EC): ethyl methyl carbonate (EMC), 3:7 by weight, water content <20 ppm; vinylene carbonate (VC), purity >99.8%, water content <100 ppm; EMC, purity 99.92%, water <6 ppm; EC, purity 99.96%, water content <2.5 ppm; fluoroethylene carbonate (FEC), purity 99.94%), Lianchuang Medicinal Chemistry Co., Ltd., China (prop-1-ene-1,3-sultone (PES), purity 98.20%), Guangzhou Tinci Co. Ltd (methylene methanedisulfonate (MMDS), purity 98.70%), Sigma Aldrich (1,3,2-dioxathiolane-2,2-dioxide (DTD), purity 98%) and TCI America (tris(trimethylsilyl) phosphite (TTSPi), purity >95%; triallyl phosphate (TAP), purity >94%). All chemicals were used as received.

Pouch cells.—200 mAh or 180 mAh NMC(442)/graphite pouch cells were obtained dry (without electrolyte) from Li-Fun Technology (Xinma Industry Zone, Golden Dragon Road, Tianyuan District, Zhuzhou City, Hunan Province, PRC, 412000). The negative to positive active mass ratio of the NMC(442)/graphite was adjusted so the cells could be cycled to 4.7 V without lithium plating. The graphite electrode of both pouch cell types consisted of 96% of 15–30 μm artificial graphite particles mixed with 2% carbon black as a conductive additive and 2% sodium carboxymethylcellulose-styrene butadiene rubber as binding agent. The positive electrode of the NMC(442)/graphite cells consisted of 96% of 5–15 μm Li[Ni_{0.4}Mn_{0.4}Co_{0.2}]O₂ (Umicore, Korea) particles mixed with 2% of carbon black and 2% of polyvinylidene fluoride (PVDF). The pouch cells were vacuum sealed in China without electrolyte and then shipped to Canada.

Pouch cell filling and formation.—The electrolyte filling, pouch sealing and formation procedure followed that described Reference 35. Degassing of the cells was performed during the first charge at 3.5 V, to remove the gas created by the reaction of electrolyte with the negative electrode and again at 4.5 V to remove the gas created by reaction of the electrolyte with the positive electrode. Degassing was performed at −90 kPa. The degassing process led to less than 5 mg of solvent loss out of 0.83 g of EC-free and 0.9 g of EC-based electrolyte. This corresponds to a maximum loss of 0.6%.

Impedance measurements.—The electrochemical impedance spectra shown in this report were measured using a BioLogic VMP3. All impedance measurements were made using a 10. mV input signal from 100. kHz to 10. mHz. The experimental setup did not allow for reproducible solution resistance measurements due to cable and connector impedance. For this reason, all impedance spectra were shifted

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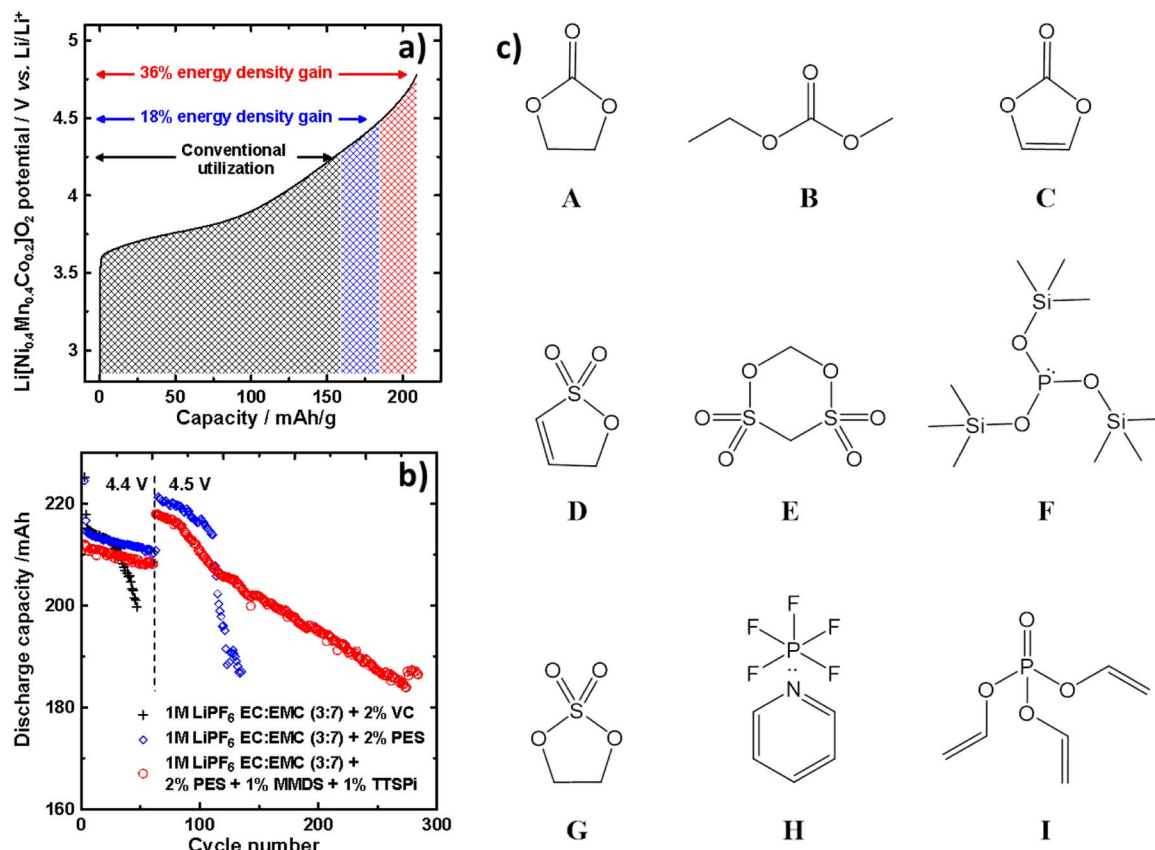


Figure 1. Potential vs. gravimetric capacity of $\text{Li}[\text{Ni}_{0.4}\text{Mn}_{0.4}\text{Co}_{0.2}]\text{O}_2$ positive electrode material (a), discharge capacity vs. cycle number of 220 mAh $\text{Li}[\text{Ni}_{0.4}\text{Mn}_{0.4}\text{Co}_{0.2}]\text{O}_2/\text{graphite}$ pouch cells cycled at 40°C at a current of $C/5$ between 2.8 – 4.4 V for the first 50 cycles and then between 2.8 – 4.5 V (b) and structures of the additives and solvents tested [A: ethylene carbonate (EC), B: ethyl methyl carbonate (EMC), C: vinylene carbonate (VC), D: propene-1,3-sultone (PES), E: methylene methanedisulfonate (MMDS), F: tris(trimethylsilyl)phosphite (TTSPi), G: 1,3,2-dioxathiolane-2,2-dioxide (DTD), H: pyridine pentafluorophosphate (PPF), I: triallyl phosphate (TAP)].

so they would start at $0\ \Omega\cdot\text{cm}^2$ on the real axis at the highest frequency measured (x-axis intercept shifted to $0\ \Omega\cdot\text{cm}^2$).

High precision coulometry.—After formation, some cells were connected to the Ultra High Precision charger (UHPC) at Dalhousie University^{36,37} to be cycled at 40°C between 2.8 – 4.5 V. The portion of the cycle between 2.8 V and 4.2 V was done at a current of $C/15$ and the portion of the cycle between 4.2 – 4.5 V was done at a current of $C/45$. This cycling protocol allows amplifying the effect of high voltage. It also represents a cycling protocol closer to what a commercial cell would experience during use. The UHPC allows the coulombic efficiency (CE) and charge-end-point capacity slippage to be measured with great accuracy and precision (30 ppm and 10 ppm respectively).³⁷

Open circuit voltage storage.—Cells were first formed following the procedure described in reference.³⁵ Cells were then put back on the charger at 40°C to be cycled twice (two full charge-discharge cycles) between 2.8 V and the upper voltage cutoff at $C/20$. Cells were then charged to the upper voltage cutoff at $C/20$ and held at that potential for 24 h. The cells were then moved to the storage station for their OCV to be monitored at $40 \pm 0.1^\circ\text{C}$ or $60 \pm 0.1^\circ\text{C}$ for 500 h. The equipment used to monitor the voltage of cells was described in an earlier publication.³⁸

Long term cycling.—Long term cycling was done using Neware battery testing stations. Cells were housed at $20^\circ\text{C} \pm 2^\circ\text{C}$ in a temperature controlled room or in a temperature controlled box at $40^\circ\text{C} \pm 0.2^\circ\text{C}$ or $55^\circ\text{C} \pm 0.2^\circ\text{C}$. Cells were cycled between 2.8 – 4.4 V using

a constant current of $C/2.5$. A constant voltage step was added at the top of charge and applied until the current dropped below $C/20$.

GC-MS measurements.—The composition of the neutral components of the electrolyte was analyzed using gas chromatography coupled with mass spectrometry. The extraction procedure of the electrolyte was described in an earlier publication.³⁹

Accelerated rate calorimetry.—The delithiated NMC442 powder and lithiated graphite powder were obtained from NMC(442)/graphite pouch cells formed to 4.5 V.⁴⁶ The single-point BET surface areas of the graphite and NMC powders were $1.5\ \text{m}^2/\text{g}$ and $0.38 \pm 0.01\ \text{m}^2/\text{g}$, respectively. ARC sample preparation followed the procedure shown in Jiang's earlier work.⁴⁰ The electrode:electrolyte mass ratios were 72 mg delithiated NMC with 23 mg electrolyte and 100 mg lithiated graphite with 100 mg electrolyte, respectively. The ARC starting temperature was set at 70°C and 50°C for NMC and graphite, respectively. ARC tests were tracked under adiabatic conditions when the sample self-heating rate (SHR) exceeded $0.03^\circ\text{C}/\text{min}$. Experiments were stopped at 350°C or when the SHR exceeded $20^\circ\text{C}/\text{min}$. To test the reproducibility of the ARC sample construction and measurements, two identical ARC samples were made and tested for every condition.

Results and Discussion

Performance of EMC-based electrolyte in NMC(442)/graphite cells cycled up to 4.4 V.—All cells used were 180 mAh NMC(442)/graphite pouch cells. The new electrolytes developed are

Table I. Volume change during the first charge to 3.5 V, 4.5 V and after cycling or storage to various upper cutoff potentials of NMC(442)/graphite cells filled with various electrolyte blends. Cycling and storage was performed at 40°C.

Electrolyte blend	Gas volume during 1st charge between 3.5 and 4.5 V /mL	Gas volume during 500 h of storage at 4.5 V and 40°C /mL	Gas volume during 500 h of cycling up to 4.5 V and 40°C /mL	Gas volume during 500 h of storage at 4.2 V and 40°C /mL	Gas volume during 500 h of storage at 4.5 V and 40°C /mL
1 M LiPF ₆ EC:EMC 3:7	1.33 ± 0.01	0.20 ± 0.01	0.06 ± 0.01	0.11 ± 0.00	0.11 ± 0.00
1 M LiPF ₆ EMC	1.77 ± 0.00	1.10 ± 0.05	0.30 ± 0.00	0.94 ± 0.01	1.04 ± 0.04
1 M LiPF ₆ EMC + 2% FEC	0.41 ± 0.05	0.25 ± 0.04	0.32 ± 0.03	0.48 ± 0.02	0.83 ± 0.00
1 M LiPF ₆ EMC + 5% FEC	0.08 ± 0.01	0.14 ± 0.01	0.01 ± 0.02	0.05 ± 0.00	0.04 ± 0.00
1 M LiPF ₆ EMC + 10% FEC	0.04 ± 0.01	0.20 ± 0.00	0.12 ± 0.01	0.08 ± 0.01	0.12 ± 0.00

compared to an EC-based electrolyte containing a ternary additive blend (2% PES + 1% DTD + 1% TTSPi). This EC-based electrolyte was chosen based on the comparison of more than 300 different additive blends.^{3,4,13,15,41–50} This extensive additive study showed that EC-based electrolytes containing this particular ternary additive blend yielded NMC-based cells with better overall performances at low (<4.25 V) or high (>4.3 V) voltage compared to other additive-containing-EC-based electrolytes. In this work, the effect of electrolyte formulation on capacity retention of cells cycled to an upper cutoff voltage of 4.4 V is shown. This upper voltage limit was chosen since many researchers reported that Li-ion cells show inferior capacity retention when tested above 4.4 V.^{2–6,50}

Ein-Eli et al.⁵¹ showed that Li could be somewhat reversibly intercalated into graphite when using a 1 M LiPF₆ EMC electrolyte. However, using EMC without a co-additive leads to cells with unmanageable gas evolution (~100% expansion, see Table I) during the first charge to 4.5 V and poor cycling efficiency due to unsatisfactory graphite passivation as will be shown later. Additives are needed to minimize the reduction of EMC and provide good graphite passivation. Electrolytes consisting of EMC and small amounts of VC are the focus of this report although other passivating agents can be used as will be shown later.

Figure 2 shows the discharge capacity (a, c) and polarization (b, d) vs. cycle number of cells containing 1 M LiPF₆ EMC:VC (98:2) electrolyte with or without pyridine pentafluorophosphate (PPF, compound H in Figure 1c) or triallyl phosphate (TAP, compound I in Figure 1c) that were cycled to 4.4 V at 20°C (a,b) and 55°C (c,d). These two co-additives (PPF and TAP) were chosen based on previous evaluation in EC-based electrolytes.^{15,52,53} Figures 2a and 2b show that cells containing EMC:VC-based electrolyte with or without PPF have very slow capacity fade and low polarization growth when cycled at 20°C and 4.4 V. This is in contrast to cells containing the EC-based electrolyte which show a more pronounced fade. Figure 2a shows, surprisingly, that cells containing an EC-free electrolyte with only 2% VC can be cycled up to 4.4 V for more than 3800 h with very limited capacity fade. By contrast, EC-based electrolytes containing 2% VC show poor performance (see Figure 1b). Figures 2a and 2b indicate that adding PPF leads to lower polarization growth in cells containing an EMC:VC electrolyte. Figures 2a and 2b also show that cells with the EMC:VC + TAP electrolyte have lower initial capacity due to a larger polarization, which eventually stabilizes as the cells cycle with little capacity loss.

Figures 2c and 2d show that cells containing an EMC:VC electrolyte without co-additives have similar capacity retention and polarization growth rate at 55°C to cells with the EC-based electrolyte containing the ternary additive blend which was the result of experiments involving thousands of cells. Figures 2c and 2d also show that adding PPF or TAP to EMC:VC electrolytes improves the high voltage-high temperature cycling performance of NMC/graphite cells. For instance cells with EMC:VC + PPF or EMC:VC + TAP electrolyte reach 80% capacity retention after 350 cycles while cells with the EC-based electrolyte reach 80% capacity after 270 cycles.

Figure 2d shows that the polarization growth rate of cells cycled at 55°C and up to 4.4 V is rather large suggesting that a large portion of the capacity loss seen in Figure 2c is from polarization growth. The addition of TAP or PPF improves the capacity retention of cells

containing an EMC:VC-based electrolyte by slowing down cell polarization growth. Figures 2a–2d clearly show that the removal of EC and the addition of small amounts of a passivating additive and co-additives can lead to a cycle life improvement of at least 30% when cycled up to 4.4 V and 55°C.

Figure 2e shows the impedance spectra of NMC(442)/graphite cells after the first charge and partial discharge to 3.8 V. Figure 2e shows that cells filled with EMC-based electrolytes can have lower impedance than cells with state-of-the-art EC-based electrolytes. Figure 2e also shows that cells with EMC:VC + TAP electrolyte have high impedance. The amount of TAP initially added to the cell remains to be optimized.

Table II shows the gas volume generated in NMC(442)/graphite cells containing various electrolyte blends during the first charge, during subsequent cycling and during storage at various voltage cut-offs and temperatures. Limiting gassing during cycling is crucial for Li-ion cells using soft packaging while limiting the gas formed during the first cycle makes cell manufacturing easier.

Aiken et al.⁵⁴ showed that certain electrolyte blends produced large amounts of gas (upwards of 50% of the initial volume of the cell) during the first charge to 3.5 V and subsequently during the first charge to 4.5 V. Table II shows that cells with EMC-based blends tested here have small gas evolution during the first charge up to 4.5 V [0.2 mL of gas is a cell volume change of 10% which is easily managed during cell manufacture]. Table II also shows that cells filled with EMC:VC-based electrolytes have very small gas production when cycled up to 4.5 V at 40°C. Even when stored at 60°C and 4.5 V for 500h, cells with EMC:VC + TAP or PPF produced small amounts of gas.

High precision coulometry and open circuit storage at 4.5 V of EMC-based electrolytes.—

Figure 3 shows the voltage vs. time of cells containing various electrolyte blends stored at 4.2 V (a-c) and 4.5 V (d-f) as well as the coulombic efficiency (CE) of cells cycled between 2.8 – 4.5 V and 40°C (g, h, i). Sinha et al. showed that the voltage drop of NMC/graphite cells during open circuit voltage is directly related to the oxidation of the electrolyte at the positive electrode. A lower voltage drop during storage indicates lower rates of electrolyte oxidation.³⁸ Burns et al. also showed in an earlier paper that cells with higher CE usually have longer cycle life.⁸ Monitoring the effect of electrolyte blends on the voltage drop during storage and CE during cycling can reveal whether changes to the electrolyte will lead to longer lifetime or not in a matter of a few weeks.

Figures 3a, 3d and 3g show why co-additives are necessary in EMC-based electrolyte. For instance, Figures 3a, 3d and 3g show that adding 2% VC to EMC leads to smaller voltage drops at both 4.2 V and 4.5 V and higher CE during cycling between 2.8 - 4.5 V compared to pure EMC electrolyte or EC-based electrolyte with a ternary additive blend. Figures 3a and 3d also indicate that EMC-based electrolytes yield greater advantage at high voltage based on the voltage drop improvement from 4.2 V to 4.5 V.

Figures 3b, 3e and 3h show that FEC can also enable the use of EMC-based electrolyte. For instance, Figures 3b, 3e and 3h show that cells with EMC and 5% or 10% FEC have lower voltage drop during storage at 4.2 V and 4.5 V as well as better CE during cycling at 4.5 V than cells filled with the EC-based electrolyte containing

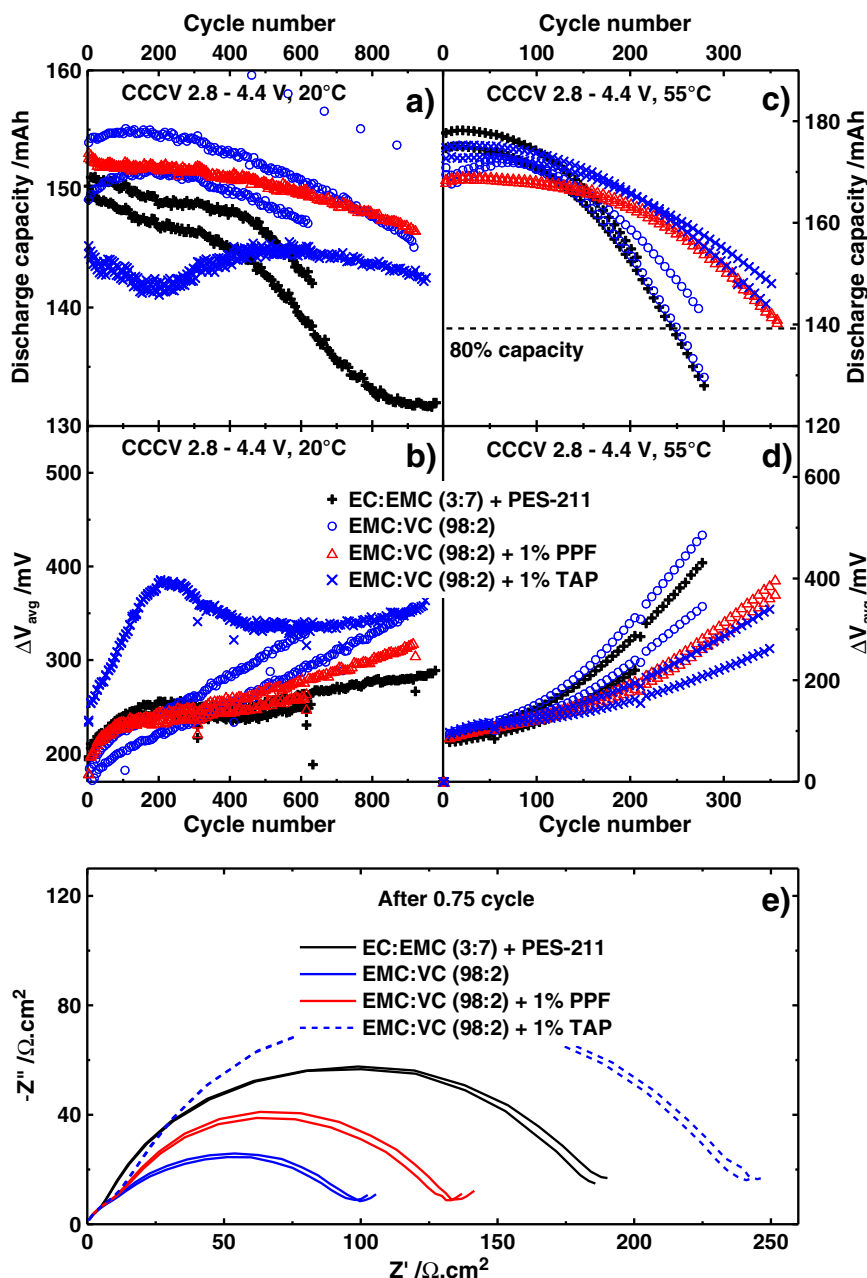


Figure 2. Discharge capacity (a) and polarization growth (b) vs. cycle number of NMC(442)/graphite cells cycled between 2.8 – 4.4 V at 20°C; discharge capacity (c) and polarization growth (d) vs. cycle number of NMC(442)/gr cells cycled between 2.8 – 4.4 V and 55°C; area normalized negative imaginary part vs. real part of the impedance of NMC(442)/graphite cells after 1 full charge to 4.4 V and a partial discharge to 3.8 V (e). In the figure caption, PES-211 refers to 2% PES + 1% DTD + 1% TTSPi (see compounds D, G and F in Figure 1c).

Table II. Volume change of 180 mAh Li[Ni_{0.4}Mn_{0.4}Co_{0.2}]O₂/graphite pouch cells during the first charge, cycling or storage at various voltages and temperatures. For comparison the cell volume before cycling is about 2 mL. PES-211 refers to 2% PES + 1% DTD + 1% TTSPi (see compounds D, G and F in Figure 1c).

Solvent blend	Gas volume during 1st charge to 3.5 V /mL	Gas volume during 1st charge between 3.5 and 4.5 V /mL	Gas volume during 500 h of storage at 4.5 V and 40°C /mL	Gas volume during 500 h of cycling up to 4.2 V and 40°C /mL	Gas volume during 500 h of cycling up to 4.4 V and 40°C /mL	Gas volume during 500 h of cycling up to 4.5 V and 60°C /mL
EC:EMC (3:7) + PES-211	0.13 ± 0.01	0.06 ± 0.00	0.08 ± 0.00	0.00 ± 0.01	−0.08 ± 0.00	0.27 ± 0.00
EMC:VC (98:2)	0.04 ± 0.01	0.22 ± 0.02	0.03 ± 0.00	−0.03 ± 0.01	−0.03 ± 0.01	0.49 ± 0.07
EMC:VC (98:2) + 1% PPF	0.24 ± 0.01	0.03 ± 0.01	0.06 ± 0.01	NA	NA	0.36 ± 0.05
EMC:VC (98:2) + 1% TAP	0.02 ± 0.01	0.23 ± 0.04	0.03 ± 0.00	0.02 ± 0.01	−0.03 ± 0.00	0.22 ± 0.02

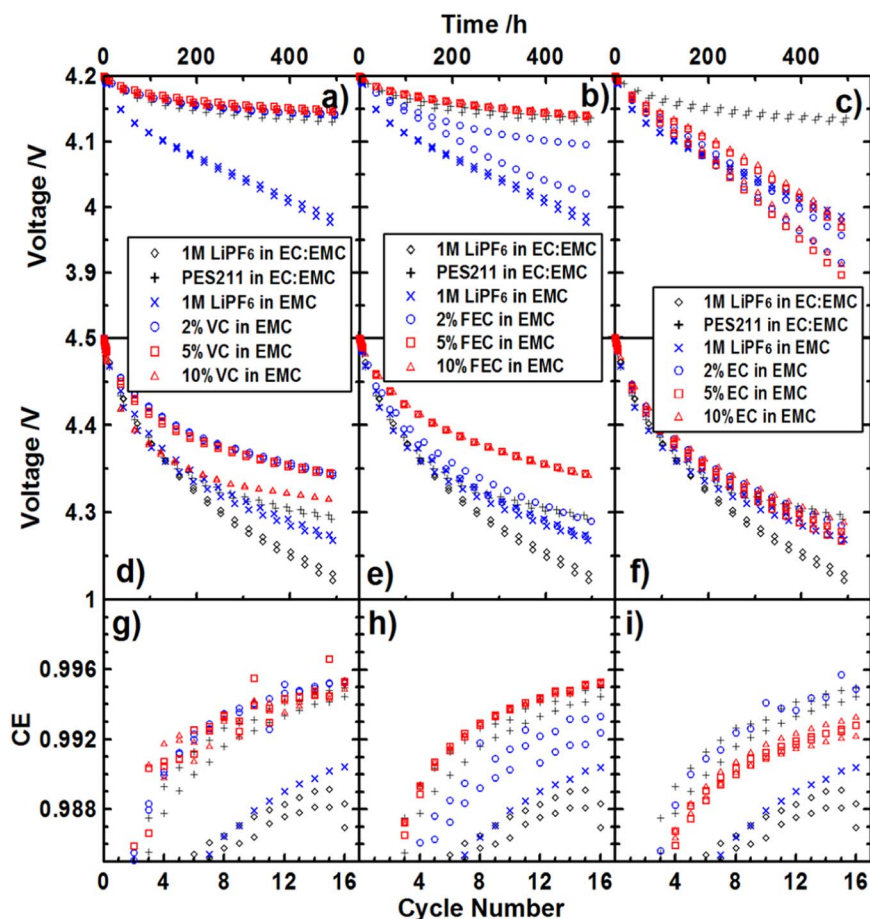


Figure 3. Voltage vs. time during open circuit voltage storage at 4.2 V (a-c) and 4.5 V (d-f) and coulombic efficiency during cycling between 2.8 – 4.5 V (g-i) for NMC(442)/graphite cells filled with various electrolyte blends. All experiments were performed at 40°C. In the figure caption, PES-211 refers to 2% PES + 1% DTD + 1% TTSPi (see compounds D, G and F in Figure 1c).

the ternary additive blend. Similarly to the EMC:VC system, the biggest improvement imparted by the EMC:FEC electrolyte seems to be at high potential. Figures 3b, 3e and 3h also indicate that there is a minimum amount of FEC needed. For instance cells with 2% FEC show marginal improvement over cells containing EC-based electrolytes. EMC:FEC electrolytes have been previously reported in the literature.^{24,32,55,56} However, these have typically high FEC loading (10–40%)⁵⁷ which will ultimately lead to high cost. Here, we show that only minimal amounts are necessary. Using large FEC loadings (10–20%) also leads to higher gas evolution during cycling and storage at elevated temperature (>40°C). For instance, Table I shows that increasing the FEC content from 5% to 10% leads to increased gas evolution during the first charge to 4.5 V, subsequent cycling and storage to 4.2 V or 4.5 V.

Figures 3c, 3f and 3i also show that lowering the EC content from 30% to 2% leads to improved high voltage performance but not low voltage (4.2 V) performance. For instance, cells filled with EMC-based electrolytes with 2% EC have to similar voltage drop during storage at 4.5 V and CE during cycling at 4.5 V as cells containing the EC-based electrolyte with the ternary additive blend. However, increasing EC content to 5% or 10% yields performance similar to cells with the EC-based blend without electrolyte additives. This clearly shows that the presence of large quantities of EC is detrimental to cells cycled to high voltage. While lowering the EC content to 2% improves cell performance, cells containing a 98EMC:2VC or 95EMC:5FEC based electrolyte show better performance.

Figure 3 clearly shows that the removal of EC leads to better cycling performance at high voltage. It also shows that the presence of small amounts of passivating agents is necessary. It also shows that the choice of passivating agent (EC, VC or FEC) has a great impact on cell performance as well. The exploratory work to be done in terms of EMC-based electrolyte optimization is considerable since multiple

compounds can enable EMC-based-EC-free electrolyte to function well at high voltage and multiple co-additives can help improve performance further.

The results presented in Figures 2 and 3 clearly show that the EC-free blends yield lower rates of solvent oxidation, better coulombic efficiency and lower polarization growth rate during high voltage cycling. Removing EC from EC-linear carbonate blends lowers the rate of electrolyte oxidation. Adding a passivating agent protects the graphite against solvent reduction and adding a co-additive (PPF and TAP in this case) lowers the rate of polarization growth of cells cycled to high voltage.

Optimization of the initial VC loading.—Burns et al.⁵⁸ and Petibon et al.⁵⁹ showed that increasing the VC loading past a certain point in EC-based electrolytes leads to cells with high impedance. Later, Petibon et al.⁶⁰ correlated this increase in cell impedance with the amount of VC consumed during cell operation. Self et al.⁶¹ and Xia et al.²⁵ showed that increased VC loading lead to higher gas generation during the first charge of NMC-based cells to high voltage and during subsequent cycling. Optimizing the VC loading is then crucial to obtain cells with good performance.

Figures 4a and 4b show the amount of VC left and the amount of EMC transesterification as a function of initial VC loading after the first cycle of cells containing 1M LiPF₆ EMC:VC electrolytes. Figure 4a shows that cells initially containing 1.7 – 3% VC have very little VC left after the first cycle. This is why cells filled with EMC:VC-based electrolytes with low initial VC loading have low gas evolution (see Table II). Figures 4a, 4b and 4d also show that at initial VC loadings of 1.7–3%, a very small amount of VC is left in the electrolyte past a cell voltage of 3.65 V. It is still unclear whether this small amount can contribute to the formation of an SEI on the positive electrode. Nonetheless, the data presented in

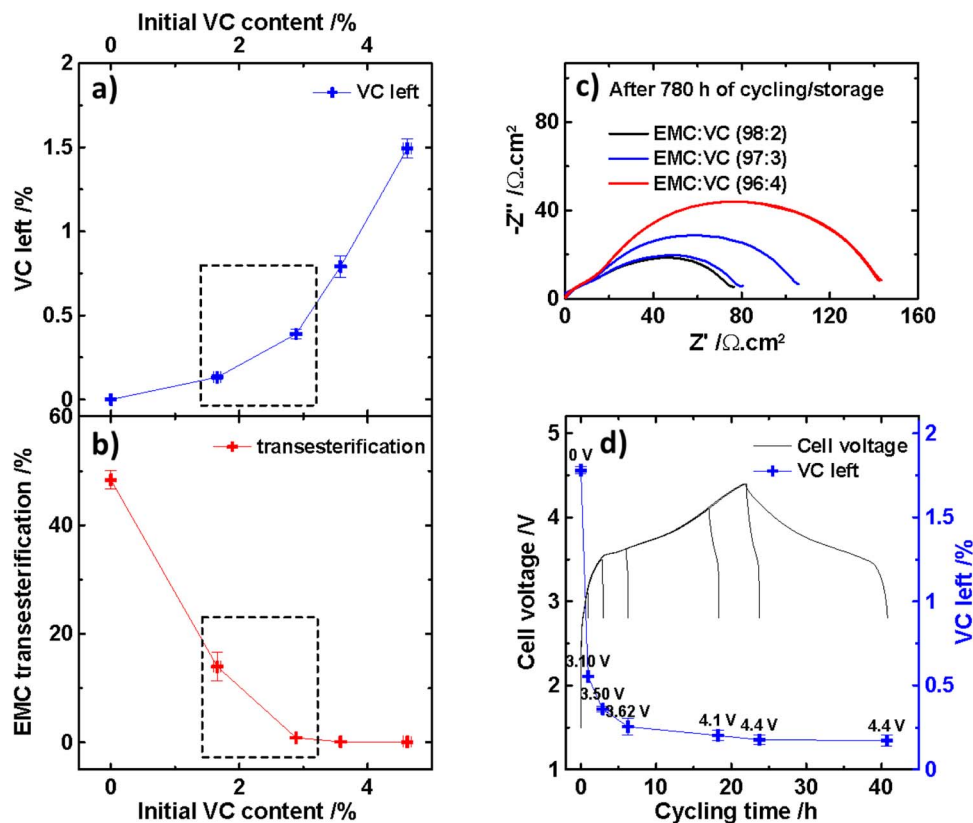


Figure 4. Amount of VC left (a) and % EMC trans-esterification (b) vs. initial VC content in NMC(442)/graphite cells that underwent one full cycle between 2.8 – 4.4 V at 40°C; area normalized negative imaginary part vs. real part of the impedance of NMC(442)/graphite cells after 780 h of cycling and storage at 4.4 V and 40°C (c) and amount of VC left vs. cycling time during the first cycle at 40°C in NMC(442)/graphite cells initially filled with an electrolyte containing 1.8% VC (d).

Figures 3 and 4 suggests that left-over VC leads to poor high voltage storage (Figure 3d), coulombic efficiency (Figure 3g) and impedance increase (Figure 4c). This strongly suggests that either VC or the SEI it creates at the positive electrode has poor oxidation stability. Figure 3i also suggests that the same is true for EC.

Figure 4b shows the amount of EMC that underwent transesterification during formation producing dimethyl carbonate (DMC) and diethyl carbonate (DEC).^{62,62–65} These transesterification reactions are catalyzed by the presence of lithium alkoxides generated by the reduction of linear carbonates.^{66–68} Figure 4b shows that EMC reduction/transesterification is mostly suppressed at an initial VC loading between 1.8 – 3%. This shows that the optimal VC loading required to passivate the graphite surface and to prevent substantial EMC reduction lies between 1.8 – 3% with the cell design used in this study.

Figure 4c shows the impedance of cells filled with EMC-based electrolytes with incremental initial VC contents, after 780 h of cycling and storage. Figure 4c shows that increasing the VC loading from 2 to 4% leads to increased cell impedance, similar to the effect of VC loading on the impedance of cells filled with EC-based electrolytes.^{58,59} Figure 4c shows that the VC loading needs to be adjusted to prepare cells with acceptable impedance.

Figure 4d shows the amount of VC remaining as a function of cycling time during the first charge to various voltage cut-offs of NMC(442)/graphite cells filled with 1 M LiPF₆ EMC:VC (98.2:1.8) electrolyte. Figure 4d shows that most of the consumption of VC occurs between 0 – 3.1 V which spans the cell voltage window where VC is reduced at the graphite surface.^{25,69} Figure 4d also shows that once the cell reaches 3.62 V, only a small amount of VC is left in the cell (~ 0.25% of the original 1.8%) which can be oxidized when the cell is charged to high potential.

Figures 4a–4d show that the optimization of the initial VC loading leads to cells with low impedance, low EMC reduction and transesterification, low gas production during the first charge to high voltage and during subsequent cycling. In the cells used and with the electrolyte volume added in this study, the optimal VC loading seems to be between 1.8 and 3%. This content would need to be adjusted in other cell designs.

Electrolyte conductivity and safety.—Figure 5a shows the conductivity of linear alkyl carbonate-based electrolytes as a function of temperature. In typical electrolyte, EC is used for its graphite passivation properties and high dielectric constant.⁷⁰ Figure 5a shows that a 1 M LiPF₆ EMC electrolyte has a conductivity ranging from 2 mS·cm^{−1} to 6 mS·cm^{−1} between −20°C and 60°C, respectively. Figure 5a also shows that the addition of EC to the solvent blend is only advantageous in terms of conductivity at temperatures higher than −10°C. For instance, 1 M LiPF₆ EMC has similar conductivity as 1 M LiPF₆ EC:EMC (3:7) below −15°C. The conductivity range of the EMC-based electrolyte presented here is acceptable for many consumer electronics and energy storage cells where high C-rate capability is not needed.

Figure 5a also shows that blending DMC with EMC improves the electrolyte conductivity. For instance adding 50% DMC gives a conductivity improvement of about 30% throughout the temperature range tested. Electrolytes of LiPF₆ dissolved in pure DMC show better conductivity than EMC or EMC:DMC mixtures at temperatures higher than 5°C. The break in the conductivity vs. temperature profile corresponds to the liquidus phase transition of the electrolyte as shown in the differential thermal analysis data shown in Figure 5b. The peaks labeled S and L in Figure 5b correspond to the temperatures of the solidus and liquidus points of the LiPF₆-DMC phase diagram at 1 M LiPF₆. 1 M LiPF₆ in DMC electrolyte would be suitable in cells

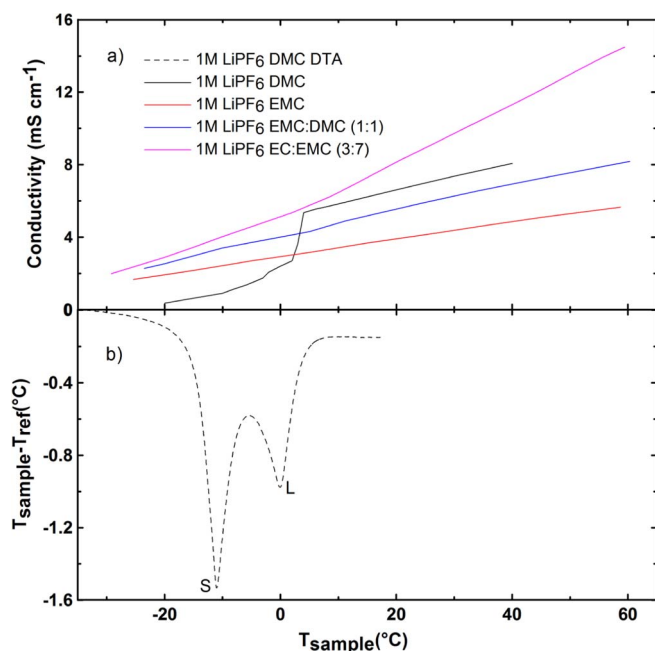


Figure 5. Conductivity vs. temperature of linear alkyl carbonate-based electrolytes (a) and differential thermal analysis data of a 1 M LiPF₆ DMC electrolyte (see Reference 71 for a description of the differential thermal analysis apparatus used). The peaks labeled S and L in Figure 5b correspond to the temperatures of the solidus and liquidus points of the LiPF₆-DMC phase diagram at 1 M LiPF₆.

used where temperatures do not go below 5–10°C and would yield cells with better rate capabilities than cells filled with an EMC-based electrolyte.

Figures 6a and 6b show the self-heating rate, measured using accelerating rate calorimetry, of lithiated graphite powder (6a) and delithiated NMC(442) powder (6b) mixed with electrolyte as a function of temperature. These measurements are very useful in evaluating the impact the electrolyte composition can have on cell safety. Self-heating events indicate the occurrence of exothermic reactions. It is important to have as large an onset temperature as possible and as low a self-heating rate as possible in order to help ensure the safety of Li-ion cells under abuse scenarios.

Figure 6a shows that lithiated graphite powder soaked in 1 M LiPF₆ EC:EMC (3:7) electrolyte has an exothermic event that peaks at 0.2°C/min with an onset around 90°C and an exothermic event around 150°C leading to thermal run-away. Figure 6a shows that the addition of the ternary additive blend leads to the suppression of the exotherm below 150°C thus improving the thermal stability of lithiated graphite. Figure 6a shows that lithiated graphite soaked in the 98% EMC + 2% VC electrolyte has an exothermic event starting at 70°C which leads to a self-heating rate of around 0.9°C/min. This reactivity at low temperature may lead to safety issues in Li-ion cells during failure events. However, Figure 6a shows that the addition of 2% TAP suppresses this exothermic event and provides similar thermal stability to EC-based electrolytes containing the ternary additive blend.

Figure 6b shows that delithiated NMC(442)/graphite powder goes into thermal runaway around 225°C in EC-based electrolyte blends (data for EC-based blends with ternary additive blends are not available). Figure 6b also shows that NMC(442) soaked in 98% EMC + 2% VC electrolyte has an exothermic event around 125°C. However Figure 6b shows once again that the addition of TAP leads to better safety properties. For instance, delithiated NMC(442)/graphite powder soaked in 1 M LiPF₆ EMC:VC + TAP shows a higher temperature thermal run-away than EC:EMC electrolyte.

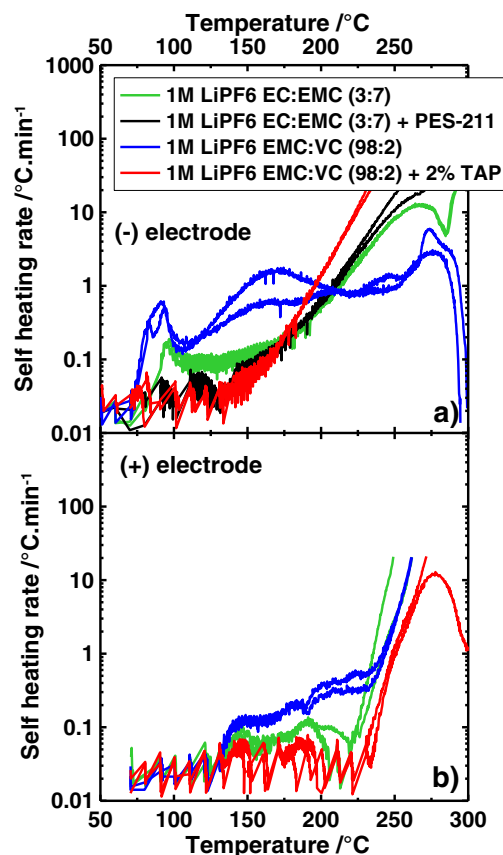


Figure 6. Self heating rate vs. temperature of lithiated graphite powder (a) and delithiated NMC(442) powder (b) recovered from NMC(442)/graphite pouch cells charged to 4.5 V and filled with various electrolyte blends.

Conclusions

Electrolytes consisting of LiPF₆ dissolved in linear carbonates and containing small amounts of passivating additives and other co-additives provide excellent cycling performance to NMC(442)/graphite cells during use at high voltage at both room and high temperatures. Such electrolytes provide cells with low gas evolution, low impedance (with the right co-additive choice), low rates of electrolyte oxidation, good graphite passivation, acceptable electrolyte conductivity and good safety (with the right co-additive). These blends are expected to be low cost.

The removal of EC has been shown to enhance high voltage performance of NMC(442)/graphite cells at both room temperature and high temperature. Several compounds can enable the use of EMC-based electrolyte such as VC, FEC, or EC (added at the additive level, see Figure 3). Adjusting the loading of these passivating agents is key in order to get cells with good overall performance. The loading of the passivating agent needs to be high enough to passivate the graphite electrode and low enough to ensure low impedance (for VC) and low gassing during cell cycling (for VC and FEC). The addition of a co-additive helps lower the polarization growth during high voltage cycling as well as helps improve safety.

These results clearly show that large amounts of EC are not needed and are actually detrimental to the cycle life of NMC/graphite cells operated to high voltage. The use of solvents with oxidation potential higher than organic carbonate is not mandatory in order to dramatically enhance high voltage cycling performance.

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